Does Continued Pretraining on a Learner Corpus Improve Automated Essay Scoring on English Proficiency Tests? Evidence from EFCAMDAT

Duy Anh Nguyen
University of Greenwich
na6367q@gre.ac.uk
anhndgcc240003@gmail.com

Abstract

Automated Essay Scoring (AES) for English proficiency assessment increasingly relies on pretrained transformer models, yet these models are typically trained on general-domain English and may under-represent second-language learner writing. This study investigates whether domain-adaptive continued pretraining (DAPT) on a learner-writing corpus improves transformer-based AES for English proficiency assessment. We perform DAPT on BERT, RoBERTa, and DistilBERT using the EFCAMDAT corpus, then compare the adapted models with their original checkpoints on two English proficiency test datasets, FCE and IELTS, in both in-domain scoring and few-shot cross-dataset transfer. Full-corpus DAPT produces mixed effects across models, datasets, and metrics. Subsequent lexical and syntactic analyses suggest mismatches between EFCAMDAT and the downstream datasets in proficiency level, genre, and communicative purpose. We therefore repeat DAPT on BERT using proficiency-specific EFCAMDAT subsets instead of the full corpus. This targeted DAPT improves downstream scoring most clearly on FCE, where DAPT on intermediate-level texts outperforms both full-corpus DAPT and the non-adapted baseline. However, these gains do not consistently transfer to the few-shot cross-dataset experiment. Overall, continued pretraining on learner writing can benefit in-domain AES when the pretraining corpus is sufficiently aligned with the target assessment setting, yet it does not automatically improve cross-test transferability.

1 Introduction

English proficiency assessment plays an important role in educational placement, certification, and language learning, with writing being a key component because it reflects a learner's ability to organize ideas and communicate effectively in English. However, evaluating writing is resource-intensive and time-consuming, especially when large numbers of responses must be assessed consistently. Therefore, Automated Essay Scoring (AES) has clear practical value in English proficiency settings, where it can support efficient and timely evaluation of learner writing.

Despite this practical importance, AES for English proficiency writing remains challenging. Unlike many general essay datasets, English assessment essays are produced by L2 learners, which often contain distinctive lexical, grammatical, and discourse patterns associated with developing language ability. Yet recent transformer-based AES approaches commonly rely on general-domain pretrained language models (Qiu et al., 2024; Schmalz and Brutti, 2021), raising the question of whether models adapted to learner writing may be better suited to this setting.

Beyond the challenge of modeling learner writing itself, AES for English proficiency assessment must also contend with variation across tests. Currently, a wide range of standardized English proficiency tests are used worldwide, such as IELTS, TOEFL, and Cambridge English examinations, each of which differs in task type, prompt design, scoring criteria, and assessment context. This diversity creates a practical challenge for AES, as a model developed for one test may not perform reliably on another. Building a separate AES system for every individual test would be both time-consuming and inefficient, especially as new test formats and writing tasks continue to emerge. A more practical direction is therefore to develop an AES approach with stronger transferability, so that one model can generalize more effectively across different English proficiency assessments.

This study investigates whether continued pretraining on an English learner corpus yields improved AES performance on English proficiency tests. We perform domain-adaptive pretraining on BERT, RoBERTa, and DistilBERT, using the EF-

CAMDAT corpus (Geertzen et al., 2014; Huang et al., 2017) as pretraining data. Our study conducts two main experiments: baseline comparisons and cross-dataset transfer, using two different datasets for downstream scoring tasks.

We hypothesize that continued pretraining on a learner corpus can improve both in-domain and cross-dataset AES performance by shifting the model's parameters away from the native-English distribution of the original checkpoint and toward the linguistic distribution of L2 learner writing represented in the downstream assessment data. Taken together, our research aims to answer the following questions:

RQ1 Does continued pretraining on a learner corpus improve AES performance on English proficiency tests compared to general-domain transformer baselines?

RQ2 Does continued pretraining on a learner corpus improve transferability when transferring from one English proficiency test dataset to another under few-shot settings?

2 Background and Related Work

Automated Essay Scoring on English proficiency tests has been the topic of research for over 20 years. Early AES studies focusing on this specific assessment setting often relied on classical machine learning techniques and feature engineering (Burstein and Chodorow, 1999; Lonsdale and Strong-Krause, 2003; Yannakoudakis et al., 2011). While these early studies demonstrated that AES could be effectively applied to learner writing, their approaches were typically developed and evaluated within relatively fixed assessment settings, where the prompts for essays are static.

Since the emergence of the Transformer architecture, AES research has shifted toward language models. In the context of English proficiency assessment, several studies have explored models such as BERT and DeBERTa for scoring learner essays (Qiu et al., 2024; Schmalz and Brutti, 2021). However, most of this work relies on general-domain transformers pretrained on corpora such as BookCorpus and Wikipedia, which consist largely of native-written text (Devlin et al., 2019). This creates a mismatch between the pretraining data and the target domain, since learner writing often contains grammatical errors, awkward phrasing, and unconventional lexical usage. As a result, general-domain language models may be less effective at capturing the characteristics of learner language, which can in turn limit AES performance and robustness.

A common approach to addressing distribution mismatch is domain adaptation. In AES, domain adaptation has been explored previously, but existing studies have focused largely on addressing the problem of prompt mismatch, where models suffer performance degradation when scoring essays written in response to unseen prompts (Phandi et al., 2015; Cao et al., 2020; Jiang et al., 2023). While prompt-level domain adaptation has proven effective in such settings, it is less suitable for AES in English proficiency tests, where there is no fixed prompt because the prompts are intentionally designed to vary across different topics and task types to effectively assess the language proficiency of the test taker. In this context, the distribution of learner language itself becomes a more appropriate target for adaptation.

A common domain adaptation approach widely used with Transformer-based models is domain-adaptive pretraining (DAPT) (Gururangan et al., 2020). In this method, a pretrained language model undergoes further pretraining on text drawn from the target domain. In the context of computer-assisted language learning, Stearns et al. (2024) applied DAPT to the BERT model using the EFCAMDAT corpus, a dataset consisting of learners' writings from different levels of proficiency. Their findings demonstrate the promise of this approach, as the adapted model achieved strong results on intrinsic evaluations. However, the study did not examine whether these gains transfer to downstream tasks through task-specific fine-tuning.

Overall, prior AES work has largely relied on general-domain pretrained models or prompt-level domain adaptation. In English proficiency testing, however, prompts vary by design, making learner-language distribution a more stable adaptation target. We therefore examine whether DAPT on learner writing improves both in-domain scoring and cross-dataset robustness.

3 Experimental Setup

3.1 Data

3.1.1 Pretraining Corpus

EFCAMDAT. EFCAMDAT is a large open-access corpus of L2 English learner writing, spanning CEFR levels A1–C1 (Huang et al., 2017; Geertzen et al., 2014). The corpus comprises texts written by learners from diverse demographic backgrounds, representing 198 nationalities. In this study, we use the cleaned version released by Shatz (2020), which removes markup-heavy texts, duplicates, non-English content, ultra-short scripts, and extreme-length outliers, resulting in 45.39 million word tokens.

3.1.2 Downstream datasets

CLC FCE Dataset. The CLC FCE dataset contains 1244 scripts written by upper-intermediate English as a Second or Other Language (ESOL) learners (Yannakoudakis et al., 2011). Each script contains two responses to the two tasks in the writing section of the First Certificate in English (FCE) exam, which asks candidates to write a letter, a report, an article, a composition, or a short story between 200 and 400 words. In this dataset, each text is labeled with two different score annotations: an overall score that the script's author achieved across both writing tasks, and a score assigned to that specific response. In this study, we use the latter score as the target variable when fine-tuning and evaluating downstream performance.

IELTS Writing Scored Essays. The IELTS Writing Scored Essays is a publicly available dataset on Kaggle[1]. This dataset was previously utilized by Liew and Tan (2025) and Qiu et al. (2024) in studies on the use of Large Language Models in AES, making it a good fit for our study. The dataset contains 1274 unique essays, covering both IELTS Writing Task 1 and Task 2, each annotated with a score ranging from 0 to 9. However, in this study, we only use IELTS Task 2 essays in the experiments. Task 1 essays were excluded because the dataset did not contain the figures or diagrams referenced in the prompts, making these responses incomplete for reliable modeling. As a result, the final dataset used in our experiments consists of 710 instances.

3.2 Preprocessing

For the IELTS dataset, samples with missing essay texts or labels were removed. Duplicate entries were also excluded, along with ultra-short essays containing fewer than 100 characters. In addition, samples containing foreign language text were also discarded to improve data consistency. The public FCE dataset uses a single holistic score ranging from 0,0 to 5,3, which later AES work often converts to a 0–20 integer scale (Gaudeau, 2025; Zhang et al., 2018). In our study, we adopt the conversion scheme of Zhang et al. (2018) to map the original FCE scores to this 0–20 scale.

3.3 Domain-adaptive Pretraining (DAPT)

In this study, we perform domain-adaptive continued pretraining on three transformer-based models: BERT (Devlin et al., 2019), RoBERTa (Liu et al., 2019), and DistilBERT (Sanh et al., 2020), using the EFCAMDAT corpus as pretraining data. We selected the models above because they have been implemented in prior AES research (Wang et al., 2022; Gui, 2025; Qiu et al., 2024), thus creating a well-established foundation for this study. BERT, in particular, has previously been used as the base model for continued pretraining on the EFCAMDAT corpus by Stearns et al. (2024). Their study reported positive gains in intrinsic evaluation relative to its original checkpoint, which motivates us to examine whether such benefits also transfer to downstream AES on English proficiency assessment essays. For the specific pretraining details, see Appendix A.

Following Stearns et al. (2024), we conduct continued pretraining with a Masked Language Modeling objective, masking 15% of WordPiece tokens in the training data and training the model to predict the masked items from their surrounding context. Their study also mentions filtering out texts with more than 512 WordPiece tokens, which we also follow.

3.4 Experiments

To answer our research questions, we conduct two main experiments: baseline comparison and cross-dataset transfer. Figures 1 and 2 visualize these experiments.

Baseline Comparison. In this experiment, the domain-adapted models are fine-tuned and evaluated separately on the FCE and IELTS datasets. Their performance is then compared with that of the corresponding non-adapted base models to determine whether DAPT on EFCAMDAT improves downstream scoring performance on the two datasets.

Cross-dataset Transfer. In this experiment, we examine downstream cross-dataset transfer learning in both directions: from IELTS to FCE and vice

[1] https://www.kaggle.com/datasets/mazlumi/ielts-writing-scored-essays-dataset

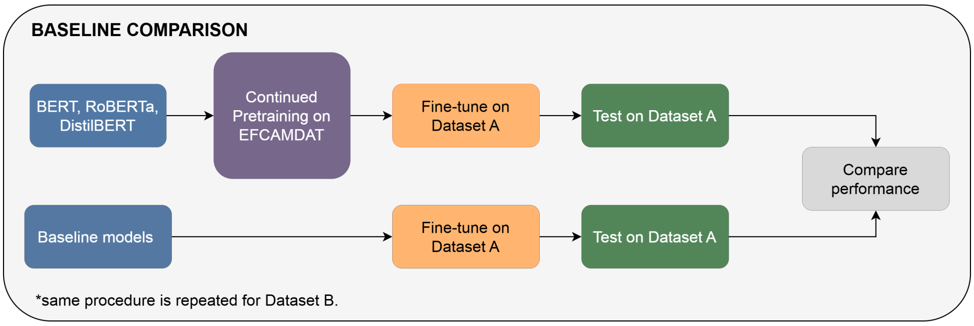

Figure 1: Baseline comparison setup. For each encoder architecture, the domain-adapted model is obtained by continuing pretraining from the corresponding base checkpoint on EFCAMDAT. The adapted and non-adapted models are then fine-tuned and evaluated on the same downstream dataset, and their performance is compared under identical fine-tuning settings. The same procedure is repeated for both downstream datasets.

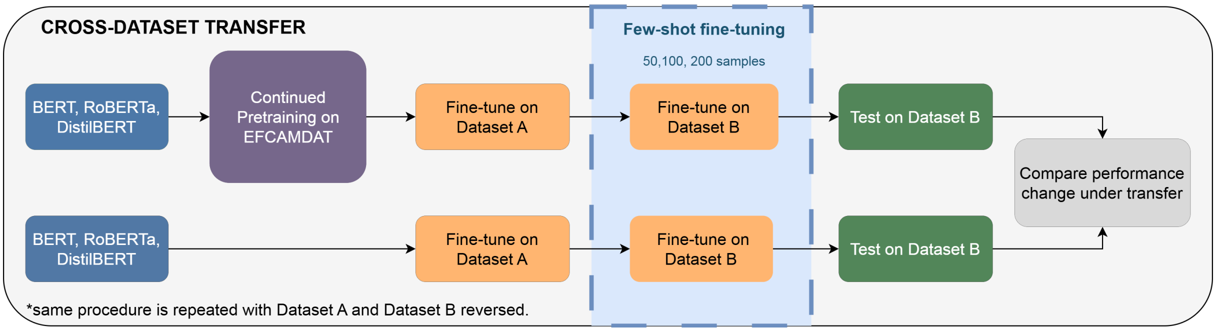

Figure 2: Cross-dataset transfer setup. For each encoder architecture, both the domain-adapted model and its corresponding non-adapted baseline are first fine-tuned on a source dataset, then adapted to the target dataset using a few-shot fine-tuning stage with 50, 100, and 200 target training samples, and finally evaluated on the target test set. Transfer-induced performance change is compared between each adapted model and its corresponding base checkpoint. The same procedure is repeated in both transfer directions.

versa. For each direction, the model is first fine-tuned on the source dataset, then subjected to an additional few-shot adaptation stage on the target dataset using 50, 100, and 200 training samples, before being evaluated on the target test set. Because the two datasets use different scoring ranges, the in-domain models are fine-tuned on normalized scores. We then compare the performance change associated with cross-dataset transfer between the domain-adapted models and their base checkpoints. This allows us to investigate whether continued pretraining on a learner corpus improves cross-dataset transferability, either by reducing performance degradation or by enabling larger gains under cross-dataset transfer.

3.5 Evaluation Protocol and Metrics

In our experiments, both downstream datasets are split with a 70/15/15 ratio for training, validating, and evaluation. For the FCE dataset in particular, we split by the provided author ID instead of individual text. This prevents essays written by the same learner from appearing in both training and evaluation sets, thereby reducing the risk of author-specific leakage and yielding a more reliable estimate of generalization. For all experiments, final performance is reported on the held-out test set that is not used during training or model selection. To ensure fairness, all experiments are conducted with the same hyperparameters and a fixed seed across runs. For the detailed hyperparameters used, see Appendix A.

We conduct our experiments in a prompt-unaware setting, meaning that no explicit prompt information is provided to the model. This allows us to better isolate the effect of domain-adaptive pretraining on the linguistic characteristics of the essays themselves, rather than on prompt-specific cues. Since continued pretraining is intended to improve the model's representation of learner writing, a prompt-unaware design provides a clearer test of whether such gains come from the essay text alone. This also provides a cleaner setting for our cross-dataset transfer experiment, where prompt mismatch can otherwise complicate the outcome.

Following prior AES work, we report RMSE, Quadratic Weighted Kappa (QWK) (Cohen, 1968), Pearson's correlation, and Spearman's correlation, because they are commonly reported metrics for this task (Phandi et al., 2015). However, Yannakoudakis and Cummins (2015) have highlighted limitations of both correlation-based measures and kappa-based agreement coefficients for automated text scoring evaluation. We therefore treat RMSE as the primary metric, which is also used by Gaudeau (2025) as the main metric for validation and model selection.

4 Experimental Results

In this section, we present and discuss the results obtained from the experiments described in Section 3.4.

4.1 Baseline Comparison

Table 1 presents the in-domain baseline comparison results. On IELTS Task 2, DAPT improves DistilBERT in terms of RMSE and QWK, but slightly lowers its correlation scores. In contrast, DAPT degrades both BERT and RoBERTa across all reported metrics, with the largest RMSE degradation observed for RoBERTa. On the FCE dataset, continued pretraining improves BERT in terms of RMSE and QWK, although its Spearman and Pearson correlations slightly decrease. For DistilBERT, DAPT leads to slightly weaker performance across all metrics, while for RoBERTa it improves QWK but worsens RMSE and correlation scores. These results suggest that continued pretraining on EFCAMDAT does not uniformly improve in-domain AES performance, and its effect differs across downstream datasets and the underlying encoder.

4.2 Cross-dataset Transfer

Table 2 summarizes the relative advantage of DAPT in the cross-dataset transfer experiment. We compute the DAPT advantage in two steps:

1. For each model and each value of n, we compute the transfer-induced performance change by comparing the n-shot cross-dataset transfer result with the corresponding in-domain result.

2. We then compare this performance change between each DAPT model and its corresponding non-DAPT baseline.

For QWK, Spearman, and Pearson, where higher values are better, larger values indicate better performance preservation or improvement. For RMSE, where the lower values are better, the sign is reversed so that positive values consistently indicate a more favorable transfer outcome. Thus, a positive DAPT advantage means that the DAPT model preserves performance better under transfer, either by suffering a smaller performance drop or by obtaining a larger transfer gain. The complete set of cross-transfer results used to compute these values is reported in Appendix B.1.

The results are inconsistent across transfer directions, metrics, and models. In the IELTS to FCE transfer direction, DAPT BERT shows mostly negative advantages, suggesting that continued pretraining does not improve transfer robustness for BERT in this direction. DAPT RoBERTa produces mixed results, with positive RMSE advantages at 100 and 200 shots, but mostly negative QWK and correlation advantages. In contrast, DAPT DistilBERT shows the clearest positive pattern in this direction, with positive advantages for RMSE and QWK across all shot sizes and positive correlation advantages in most cases.

The reversed direction, with FCE as the source dataset, shows a different pattern. DAPT BERT again provides little evidence of transfer improvement, with negative advantages across most metrics and shot sizes. DAPT RoBERTa performs comparatively better in this direction, showing positive RMSE advantages across all shot sizes and positive advantages for several QWK and Spearman scores. However, this improvement is not uniform, as some correlation values remain negative. DAPT DistilBERT, which showed the strongest pattern in the IELTS to FCE direction, performs consistently worse than its base counterpart in the reverse direction, with negative advantages across all reported metrics and shot sizes. Overall, these results suggest that learner-domain DAPT does not consistently improve cross-test transferability, and its effect depends strongly on the transfer direction, model architecture, and evaluation metric.

5 Discussion and Analysis

We hypothesize that the mixed effects of DAPT on the full pretraining corpus are partly attributable to mismatches between the EFCAMDAT corpus and the downstream datasets, particularly in terms of proficiency profile, genre, and communicative pur-

IELTS Task 2

Model	RMSE	QWK	Sp.	Pe.
BERT	**0.632**	**0.712**	**0.784**	0.785
DAPT DistilBERT	0.691	0.705	0.725	0.730
RoBERTa	0.706	0.661	0.783	**0.809**
DistilBERT	0.713	0.700	0.732	0.737
DAPT BERT	0.737	0.689	0.764	0.778
DAPT RoBERTa	0.852	0.489	0.739	0.766

FCE

Model	RMSE	QWK	Sp.	Pe.
DistilBERT	**2.080**	**0.614**	0.667	0.675
DAPT BERT	2.101	0.598	0.652	0.678
DAPT DistilBERT	2.115	0.587	0.639	0.665
RoBERTa	2.138	0.513	**0.686**	**0.695**
DAPT RoBERTa	2.200	0.554	0.667	0.683
BERT	2.316	0.571	0.677	0.684

Table 1: In-domain AES performance on IELTS Task 2 and FCE. Sp.=Spearman and Pe.=Pearson. DAPT models denote the corresponding base encoder further pretrained on EFCAMDAT. Best performance achieved on each dataset is shown in bold.

IELTS → FCE

Model	n	RMSE	QWK	Sp.	Pe.
BERT	50	-0.369	-0.015	-0.013	-0.012
	100	-0.395	-0.058	**0.031**	**0.025**
	200	-0.270	-0.103	**0.005**	-0.012
RoBERTa	50	-0.169	-0.206	-0.140	-0.153
	100	**0.100**	-0.010	**0.039**	**0.027**
	200	**0.191**	-0.096	-0.006	-0.020
DistilBERT	50	**0.062**	**0.031**	-0.001	**0.017**
	100	**0.045**	**0.048**	**0.024**	**0.037**
	200	**0.049**	**0.057**	**0.022**	**0.024**

FCE → IELTS

Model	n	RMSE	QWK	Sp.	Pe.
BERT	50	-0.066	-0.324	-0.032	-0.043
	100	**0.010**	-0.082	-0.020	-0.023
	200	-0.025	-0.125	-0.025	-0.040
RoBERTa	50	**0.144**	**0.096**	**0.023**	-0.011
	100	**0.113**	-0.054	-0.006	-0.025
	200	**0.012**	**0.069**	**0.018**	**0.005**
DistilBERT	50	-0.068	-0.083	-0.055	-0.053
	100	-0.067	-0.044	-0.037	-0.049
	200	-0.053	-0.043	-0.028	-0.033

Table 2: DAPT advantage in few-shot cross-test transfer. Each value compares the transfer-induced performance change of a DAPT model against its corresponding base model. Positive values indicate a more favorable transfer outcome for DAPT. RMSE is treated inversely. Sp.=Spearman; Pe.=Pearson.

pose. This hypothesis is based on the known proficiency imbalance of EFCAMDAT, which is skewed toward lower-level A1–A2 learner writing (Stearns et al., 2024), as well as our preliminary qualitative inspection of corpus prompts and sample texts. To investigate this hypothesis, we conduct quantitative analyses between CEFR proficiency subsets of the EFCAMDAT and the downstream datasets, particularly in terms of vocabulary distribution and syntactic complexity.

5.1 Vocabulary Distribution

To examine whether the pretraining corpus is lexically aligned with the downstream AES datasets, we compare vocabulary distributions across EFCAMDAT proficiency subsets and the downstream datasets. Specifically, we compute Jensen–Shannon divergence (JSD) (Lin, 1991), a symmetric divergence measure, between each CEFR-based EFCAMDAT subset and each downstream dataset. Following its use in NLP corpus comparison tasks (Lu et al., 2020), we represent each corpus as a vocabulary probability distribution, where each token is assigned a probability based on its normalized frequency in that corpus.

$$\mathrm{JSD}(P \,\|\, Q) = \frac{1}{2} D_{\mathrm{KL}}\left(P \,\middle\|\, \frac{P+Q}{2}\right) + \frac{1}{2} D_{\mathrm{KL}}\left(Q \,\middle\|\, \frac{P+Q}{2}\right). \tag{1}$$

Here, P and Q denote the vocabulary probability distributions of two corpora, and D_{KL} denotes Kullback–Leibler divergence (Kullback and Leibler, 1951). JSD measures how much each of the distributions diverges from their average distribution, $\frac{P+Q}{2}$, with lower values indicating greater lexical similarity between the two corpora.

The computed JSD results are reported in Table 3. Overall, the values suggest that the FCE dataset is lexically closest to the B1 and B2 CEFR subsets of EFCAMDAT, indicating relatively stronger vocabulary distribution alignment with intermediate learner texts. In contrast, the IELTS dataset shows the lowest divergence from the C1 subset, suggesting closer lexical alignment with more advanced learner writing.

To provide a descriptive view of the lexical differences underlying the JSD results, we extract the 20 most frequent tokens in EFCAMDAT, IELTS, and FCE, both with and without stopwords (see

CEFR subset	FCE	IELTS
A1	0.3006	0.4322
A2	0.1783	0.3398
B1	0.1373	0.2235
B2	**0.1397**	0.1810
C1	0.1575	**0.1574**

Table 3: Jensen–Shannon divergence between each EFCAMDAT CEFR proficiency subset and each downstream AES dataset. Lower values indicate greater vocabulary distribution similarity. The lowest JSD for each downstream dataset is shown in bold.

Appendix B.2 for the full list). When stopwords are included, EFCAMDAT and FCE show greater overlap in conversational high-frequency tokens, particularly pronouns such as *i*, *my*, and *you*. By contrast, IELTS contains fewer personal pronouns among its most frequent tokens and includes more general argumentative tokens, such as *people*, *their*, and *they*. The stopword-removed comparison further illustrates this pattern: EFCAMDAT and FCE contain more everyday and interactional lexical items, such as *like*, *good*, *think*, and *want*, whereas IELTS contains more topic-general and argumentative vocabulary, including *society*, *countries*, *students*, *however*, and *example*. Overall, these patterns suggest a noticeable mismatch in genre and communicative purpose between EFCAMDAT and IELTS, whereas the mismatch between EFCAMDAT and FCE appears comparatively smaller.

5.2 Syntactic Complexity

To examine syntactic alignment between the pretraining corpus and the downstream datasets, we use NeoSCA[2], a modern implementation of the L2 Syntactic Complexity Analyzer (L2SCA) (Lu, 2010) to extract syntactic complexity features from each dataset. NeoSCA is applied at the text level, and the resulting feature values are then aggregated to obtain corpus-level syntactic profiles. For the downstream datasets, we apply NeoSCA to all available texts. For the EFCAMDAT proficiency subsets, however, applying NeoSCA to the full corpus would be computationally expensive due to the large size of the dataset. Therefore, we randomly sample 500 texts from each CEFR proficiency subset and apply the same extraction procedure to each sample. Although NeoSCA outputs 14 syntactic complexity indices, we report a focused shortlist of seven features most relevant to our mismatch hypothesis, which capture differences in production-unit length, clausal density, subordination, coordination, and nominal/phrasal complexity. The results are available in Table 4.

Overall, the syntactic patterns are broadly consistent with the earlier vocabulary distribution results in Section 5.1. Among the EFCAMDAT proficiency subsets, FCE appears most closely aligned with the B1 and B2 subsets, suggesting that its syntactic profile is closer to intermediate-level learner writing. IELTS, by contrast, is closest to the C1 subset, reflecting its generally higher syntactic complexity. However, the gap between IELTS and even the closest EFCAMDAT subset remains substantial, indicating that IELTS is still relatively misaligned with the available EFCAMDAT proficiency levels. This supports the view that dataset mismatch is not only lexical, but also reflected in syntactic complexity.

6 Ablation Study

In this section, we conduct a proficiency-based ablation study, in which continued pretraining is performed on selected CEFR proficiency subsets of EFCAMDAT rather than on the full corpus. This allows us to test whether our mismatch hypothesis is correct, in which case, targeted DAPT on better-aligned learner writing can lead to stronger downstream AES performance than full-corpus DAPT. Due to the substantial computational cost of repeating domain-adaptive pretraining across all three encoder models, the proficiency-based ablation is conducted using BERT only. This choice allows us to isolate the effect of pretraining-corpus proficiency alignment while keeping the ablation computationally feasible and consistent with the main DAPT setup. In addition, because the number of C1-level texts in EFCAMDAT is limited, we combine B2 and C1 texts to form the highest-proficiency subset. This grouping provides a more viable amount of pretraining data while still representing the upper-proficiency range of the learner corpus.

Table 5 presents the results of the proficiency-based ablation. Overall, the results are consistent with the lexical and syntactic alignment analyses reported in Sections 5.1 and 5.2. For FCE, DAPT on the B1–B2 subset achieves the strongest overall performance, obtaining the lowest RMSE and highest QWK among all variants, and outperforming DAPT on the full EFCAMDAT corpus. This is consistent with the earlier observation that FCE is most

[2]https://github.com/tanloong/neosca

Source	MLT	MLC	C/T	DC/C	CT/T	CP/T	CN/T
EFCAMDAT A1	9.006	8.259	1.115	0.070	0.087	0.285	0.626
EFCAMDAT A2	10.334	8.469	1.227	0.142	0.183	0.265	0.745
EFCAMDAT B1	13.655	9.649	1.434	0.249	0.319	0.302	1.227
EFCAMDAT B2	14.408	9.758	1.537	0.297	0.384	0.326	1.412
EFCAMDAT C1	15.827	10.554	1.547	0.307	0.400	0.374	1.692
FCE train	14.227	8.299	1.729	0.351	0.467	0.222	1.208
IELTS train	20.726	11.036	1.915	0.412	0.543	0.552	2.620

Table 4: Syntactic complexity profiles of EFCAMDAT proficiency subsets and downstream training datasets.

Note. MLT = mean length of T-unit; MLC = mean length of clause; C/T = clauses per T-unit; DC/C = dependent clauses per clause; CT/T = complex T-units per T-unit; CP/T = coordinate phrases per T-unit; CN/T = complex nominals per T-unit. A T-unit refers to an independent clause together with any dependent clauses attached to it.

closely aligned with the B1–B2 EFCAMDAT subset in terms of vocabulary distribution and syntactic complexity. For IELTS, the best-performing DAPT variant is the B2–C1 subset. However, this variant still does not outperform the non-adapted BERT checkpoint, which remains strongest across RMSE and QWK. This result also aligns with earlier analysis in Section 5.2, where the C1 EFCAMDAT texts were found to be closest to IELTS, yet still different by a substantial margin.

We further extend the proficiency-based ablation to the cross-dataset transfer experiment. However, unlike the downstream scoring results, the transfer results remain inconsistent across transfer directions, metrics, and pretraining subsets (results available in Appendix B.3). In other words, restricting DAPT to proficiency-aligned EFCAMDAT subsets does not consistently reduce the transfer gap between FCE and IELTS. This suggests that proficiency alignment alone is not sufficient to improve cross-dataset transferability, especially when the two downstream datasets differ in scoring scale, task design, genre, and communicative purpose.

Taken together, these findings suggest that DAPT on a learner-writing corpus can improve downstream AES performance, but its benefits are conditional on the alignment between the pretraining corpus and the target dataset. The proficiency-based ablation supports this interpretation for downstream scoring, but the same pattern does not consistently extend to cross-dataset transfer, where results remain unstable across transfer directions, metrics, and pretraining subsets.

Pretraining Data	**RMSE**	**QWK**	**Sp.**	**Pe.**
FCE				
B1–B2 only	**2.064**	**0.609**	0.662	0.679
Full EFCAMDAT	2.101	0.598	0.652	0.678
B2–C1 only	2.143	0.550	0.636	0.653
A1–A2 only	2.177	0.581	0.647	0.665
No EFCAMDAT	2.316	0.571	**0.677**	**0.684**
IELTS				
No EFCAMDAT	**0.632**	**0.712**	0.784	0.785
B2–C1 only	0.695	0.627	0.725	0.751
A1–A2 only	0.735	0.687	**0.795**	**0.793**
Full EFCAMDAT	0.737	0.689	0.764	0.778
B1–B2 only	0.758	0.663	0.772	0.777

Table 5: Ablation results on FCE and IELTS. Within each dataset, rows are sorted by RMSE in ascending order. "No EFCAMDAT" denotes the base BERT model checkpoint without DAPT. Sp.=Spearman and Pe.=Pearson. Best result for each dataset and metric is shown in bold.

7 Conclusion

This study set out to examine whether learner-domain DAPT provides a reliable benefit for automated essay scoring on English proficiency tests. Rather than finding a uniform improvement from continued pretraining on EFCAMDAT, our results show a more conditional pattern. Full-corpus DAPT produced mixed effects across datasets, models, and metrics, suggesting that exposure to a broad learner corpus alone is not sufficient to guarantee better AES performance. Further lexical and syntactic analyses, followed by proficiency-based ablations, indicate that DAPT can benefit AES on learners' writings, but only when the pretraining data is sufficiently aligned with the downstream assessment dataset. Additionally, this effect is mainly observed in downstream scoring performance and does not consistently extend to the cross-dataset transfer setting, where DAPT does not reliably improve transferability across assessment contexts.

Limitations

This study has several limitations. First, continued pretraining was conducted using only one learner corpus, EFCAMDAT. Although EFCAMDAT is large and well suited to learner-language adaptation, its proficiency distribution, task types, and communicative purposes may not represent learner writing more broadly. As our lexical and syntactic analyses show, the corpus is not equally aligned with the downstream assessment datasets, which limits the generalizability of the observed DAPT effects.

Second, the downstream evaluation is limited to two English proficiency test datasets, FCE and IELTS Task 2. The IELTS subset is relatively small after preprocessing, and the findings may therefore be sensitive to dataset composition and splits. Future work should evaluate learner-domain DAPT on additional proficiency tests and larger writing datasets.

Third, although the main experiments include BERT, RoBERTa, and DistilBERT, the proficiency-based ablation study is conducted only with BERT due to computational cost. Therefore, the alignment-based interpretation is most directly supported for BERT, and further work is needed to determine whether the same pattern holds across other pretrained encoder architectures.

Ethics Statement

We make the code publicly available at `https://github.com/KatoTheFluffyWolf/DAPT-EFCAMDAT`. However, access to EFCAMDAT is governed by a License Agreement and is restricted to academic, non-commercial research, with conditions protecting copyright. For this reason, we will not redistribute the corpus or any text-derived data artifacts with our code, including the model checkpoints, as it risks the regulated terms being violated beyond our control. Researchers wishing to reproduce our experiments must obtain access to EFCAMDAT through the official provider. At the same time, we believe that making the trained models available would benefit future research by reducing computational cost and enabling more direct comparison with follow-up studies. We therefore plan to contact the EFCAMDAT providers to discuss whether the trained model checkpoints can be made available through the official EFCAMDAT website or another provider-approved access channel. Any such release would be subject to the providers' approval and to the same access, copyright, and non-commercial research conditions governing the corpus.

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A Training Details

In this appendix, we provide additional training details to support the reproducibility of our experiments. Specifically, Table 6 reports the key hyperparameters used during domain-adaptive continued pretraining and downstream fine-tuning, including batch size, learning rate, training duration, validation split, masking probability, and model selection settings.

Additionally, we report the validation MLM loss curves observed during continued pretraining to show that the adapted models learned from EFCAMDAT and that training proceeded without evidence of validation-loss divergence. The decision to train for only one epoch is supported by these curves: for all models, validation MLM loss decreases substantially early in training and begins to plateau toward the end of the epoch, suggesting that further training would likely yield diminishing returns. Figure 3 visualizes these learning curves.

B Supplementary Experimental Result

B.1 Cross-dataset Transfer Full Results

Table 7 reports the complete results of the cross-dataset transfer experiment described in Section 3.4. These results are used to compute the DAPT advantage values summarized in Table 2.

B.2 Top-k words

Table 8 reports the 20 most frequent tokens in EFCAMDAT, IELTS, and FCE, both before and after stopword removal. These token lists provide a descriptive view of the vocabulary patterns discussed in Section 5.1.

B.3 Ablated Cross-dataset Transfer Results

In this appendix, we report the results of the ablated cross-dataset transfer experiment discussed in Section 6. Table 9 presents the DAPT advantage values for each proficiency-based DAPT variant, while Table 10 reports the complete raw transfer results from which these values are derived. DAPT advantage is computed using the same procedure described in Section 4.2.

Overall, the ablated transfer results show that proficiency-targeted DAPT does not consistently improve cross-dataset transfer. In the IELTS to FCE direction, most DAPT advantage values remain negative across pretraining subsets and shot sizes, indicating that restricting DAPT to better-aligned EFCAMDAT subsets does not reliably reduce the transfer gap. The strongest exception is the B2–C1 subset at 200 shots, which shows small positive advantages for QWK, Spearman, and Pearson. In the opposite transfer direction, the results are somewhat more favorable, especially at 100 and 200 shots, where A1–A2, B1–B2, and B2–C1 DAPT produce positive RMSE advantages. However, the gains remain inconsistent across metrics, with B2–C1 showing the clearest positive pattern but not uniformly improving all metrics. These results suggest that proficiency alignment alone is insufficient to guarantee improved cross-test transferability.

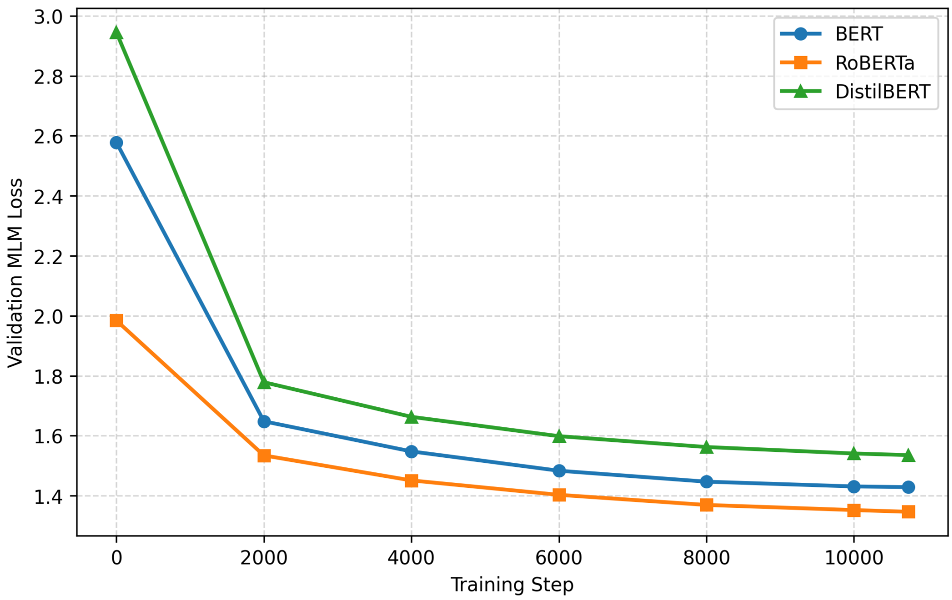

Figure 3: Validation MLM loss of BERT, RoBERTa, and DistilBERT during continued pretraining on EFCAMDAT. Lower values indicate better prediction of masked WordPiece tokens on the held-out validation split. At step 0, the reported MLM loss is that of the corresponding base checkpoint without DAPT.

Stage	Hyperparameter / Setting	Value
Continued pretraining	Total training steps	10737
	Hardware	NVIDIA L4 GPU
	Random seed	42
	Maximum sequence length	512
	MLM masking probability	0.15
	Training epochs	1
	Batch configuration	Train BS = 4, eval BS = 8, grad. accum. = 16, effective train BS = 64
	Learning rate	5×10^{-5}
	Optimizer	AdamW
	Weight decay	0.01
	Warm-up	6% of total training steps
	Learning rate scheduler	Linear
	Precision	FP16
	Validation split	5%
Downstream fine-tuning	Maximum sequence length	512
	Learning rate	2×10^{-5}
	Batch configuration	Train BS = 8, eval BS = 16
	Training epochs	5
	Model selection metric	RMSE

Table 6: Main hyperparameters used for continued pretraining and downstream fine-tuning.

Table 7: Full cross-test transfer results. Values in parentheses show the difference from the corresponding model's in-domain result.

Model	n	RMSE ↓	QWK ↑	Spearman ↑	Pearson ↑
IELTS → FCE					
DAPT BERT	In-domain	2.101	0.598	0.652	0.678
	50	2.557 (+0.456)	0.429 (-0.169)	0.490 (-0.162)	0.515 (-0.163)
	100	2.433 (+0.332)	0.483 (-0.115)	0.596 (-0.056)	0.616 (-0.062)
	200	2.271 (+0.170)	0.510 (-0.088)	0.590 (-0.062)	0.605 (-0.073)
BERT	In-domain	2.316	0.571	0.677	0.684
	50	2.403 (+0.087)	0.417 (-0.154)	0.528 (-0.149)	0.533 (-0.151)
	100	2.253 (-0.063)	0.514 (-0.057)	0.590 (-0.087)	0.597 (-0.087)
	200	2.216 (-0.100)	0.586 (+0.015)	0.610 (-0.067)	0.623 (-0.061)
DAPT RoBERTa	In-domain	2.200	0.554	0.667	0.683
	50	2.671 (+0.471)	0.265 (-0.289)	0.485 (-0.182)	0.487 (-0.196)
	100	2.233 (+0.033)	0.517 (-0.037)	0.661 (-0.006)	0.664 (-0.019)
	200	2.196 (-0.004)	0.495 (-0.059)	0.637 (-0.030)	0.640 (-0.043)
RoBERTa	In-domain	2.138	0.513	0.686	0.695
	50	2.440 (+0.302)	0.430 (-0.083)	0.644 (-0.042)	0.652 (-0.043)
	100	2.271 (+0.133)	0.486 (-0.027)	0.641 (-0.045)	0.649 (-0.046)
	200	2.325 (+0.187)	0.550 (+0.037)	0.662 (-0.024)	0.672 (-0.023)
DAPT DistilBERT	In-domain	2.115	0.587	0.639	0.665
	50	2.395 (+0.280)	0.432 (-0.155)	0.493 (-0.146)	0.525 (-0.140)
	100	2.297 (+0.182)	0.530 (-0.057)	0.570 (-0.069)	0.602 (-0.063)
	200	2.208 (+0.093)	0.589 (+0.002)	0.590 (-0.049)	0.626 (-0.039)
DistilBERT	In-domain	2.080	0.614	0.667	0.675
	50	2.422 (+0.342)	0.428 (-0.186)	0.522 (-0.145)	0.518 (-0.157)
	100	2.307 (+0.227)	0.509 (-0.105)	0.574 (-0.093)	0.575 (-0.100)
	200	2.222 (+0.142)	0.559 (-0.055)	0.596 (-0.071)	0.612 (-0.063)
FCE → IELTS					
DAPT BERT	In-domain	0.737	0.689	0.764	0.778
	50	0.832 (+0.095)	0.375 (-0.314)	0.699 (-0.065)	0.699 (-0.079)
	100	0.743 (+0.006)	0.628 (-0.061)	0.727 (-0.037)	0.742 (-0.036)
	200	0.737 (+0.000)	0.660 (-0.029)	0.748 (-0.016)	0.755 (-0.023)
BERT	In-domain	0.632	0.712	0.784	0.785
	50	0.661 (+0.029)	0.722 (+0.010)	0.751 (-0.033)	0.749 (-0.036)
	100	0.648 (+0.016)	0.733 (+0.021)	0.767 (-0.017)	0.772 (-0.013)
	200	0.607 (-0.025)	0.808 (+0.096)	0.793 (+0.009)	0.802 (+0.017)
DAPT-RoBERTa	In-domain	0.852	0.489	0.739	0.766
	50	0.717 (-0.135)	0.600 (+0.111)	0.754 (+0.015)	0.734 (-0.032)
	100	0.792 (-0.060)	0.477 (-0.012)	0.723 (-0.016)	0.733 (-0.033)
	200	0.726 (-0.126)	0.661 (+0.172)	0.760 (+0.021)	0.769 (+0.003)
RoBERTa	In-domain	0.706	0.661	0.783	0.809
	50	0.715 (+0.009)	0.676 (+0.015)	0.775 (-0.008)	0.788 (-0.021)
	100	0.759 (+0.053)	0.703 (+0.042)	0.773 (-0.010)	0.801 (-0.008)
	200	0.592 (-0.114)	0.764 (+0.103)	0.786 (+0.003)	0.807 (-0.002)
DAPT-DistilBERT	In-domain	0.691	0.705	0.725	0.730
	50	0.822 (+0.131)	0.547 (-0.158)	0.666 (-0.059)	0.669 (-0.061)
	100	0.726 (+0.035)	0.623 (-0.082)	0.707 (-0.018)	0.693 (-0.037)
	200	0.687 (-0.004)	0.669 (-0.036)	0.716 (-0.009)	0.725 (-0.005)
DistilBERT	In-domain	0.713	0.700	0.732	0.737
	50	0.776 (+0.063)	0.625 (-0.075)	0.728 (-0.004)	0.729 (-0.008)
	100	0.681 (-0.032)	0.662 (-0.038)	0.751 (+0.019)	0.749 (+0.012)
	200	0.656 (-0.057)	0.707 (+0.007)	0.751 (+0.019)	0.765 (+0.028)

For RMSE, lower is better; therefore, a positive difference from in-domain means performance worsened, while a negative difference means performance improved. For QWK, Spearman, and Pearson, higher is better.

Rank	EFCAMDAT		IELTS		FCE	
	Token	Count	Token	Count	Token	Count
Panel A: All tokens						
1	the	1,915,133	the	7,742	the	14,921
2	i	1,840,818	to	5,825	i	13,789
3	and	1,518,481	of	4,445	to	12,429
4	a	1,160,866	and	4,150	and	8,029
5	to	1,141,105	in	3,665	a	5,902
6	in	986,490	a	3,236	in	5,739
7	my	918,668	is	2,773	you	5,691
8	is	915,013	that	2,560	of	5,320
9	you	588,095	for	1,986	it	4,780
10	of	557,787	are	1,681	was	4,086
11	are	404,345	it	1,672	that	4,023
12	have	386,269	people	1,672	is	3,899
13	for	384,402	be	1,615	for	3,638
14	at	341,735	their	1,452	my	3,196
15	i'm	312,911	this	1,407	have	3,163
16	it	298,756	as	1,334	we	2,643
17	like	285,750	they	1,215	be	2,408
18	very	284,760	can	1,116	at	2,377
19	that	279,017	not	1,037	would	2,140
20	with	275,151	have	1,003	but	2,091
Panel B: Non-stopword tokens						
1	like	285,750	people	1,672	would	2,140
2	work	159,293	many	529	like	1,962
3	people	158,429	also	472	show	1,388
4	good	142,338	one	443	time	1,159
5	years	125,333	would	426	people	1,095
6	one	109,100	children	413	dear	974
7	day	108,973	life	396	think	964
8	live	108,392	time	396	good	940
9	job	103,874	could	358	could	869
10	lot	97,251	believe	354	money	858
11	time	97,084	students	342	one	838
12	old	92,461	however	341	know	836
13	name	91,261	countries	330	shopping	766
14	city	91,138	society	301	life	765
15	going	88,269	example	296	first	756
16	many	85,801	work	284	also	735
17	think	84,975	may	283	clothes	658
18	get	84,087	world	282	really	618
19	two	83,406	money	263	much	612
20	love	82,810	country	256	want	591

Table 8: Top 20 most frequent tokens in EFCAMDAT, IELTS, and FCE, reported with and without stopwords.

IELTS → FCE

Pretraining data	n	RMSE	QWK	Sp.	Pe.
Full EFCAMDAT	50	-0.369	-0.015	-0.013	-0.012
	100	-0.395	-0.058	**0.031**	**0.025**
	200	-0.270	-0.103	**0.005**	-0.012
A1–A2 only	50	-0.238	-0.105	-0.040	-0.057
	100	-0.276	-0.022	**0.014**	**0.017**
	200	-0.222	-0.087	-0.011	-0.025
B1–B2 only	50	-0.338	-0.072	-0.039	-0.052
	100	-0.387	-0.137	-0.034	-0.046
	200	-0.330	-0.082	**0.023**	**0.010**
B2–C1 only	50	-0.242	-0.008	-0.022	-0.021
	100	-0.409	-0.015	**0.017**	**0.024**
	200	-0.149	**0.002**	**0.050**	**0.040**

FCE → IELTS

Pretraining data	n	RMSE	QWK	Sp.	Pe.
Full EFCAMDAT	50	-0.066	-0.324	-0.032	-0.043
	100	**0.010**	-0.082	-0.020	-0.023
	200	-0.025	-0.125	-0.025	-0.040
A1–A2 only	50	-0.052	-0.242	-0.064	-0.065
	100	**0.106**	**0.015**	-0.026	-0.014
	200	**0.062**	-0.062	-0.068	-0.045
B1–B2 only	50	-0.040	-0.313	-0.061	-0.062
	100	**0.068**	-0.004	-0.008	-0.020
	200	**0.073**	-0.055	-0.035	-0.032
B2–C1 only	50	-0.088	-0.232	0.000	-0.025
	100	**0.051**	**0.067**	**0.045**	**0.034**
	200	**0.018**	**0.039**	**0.011**	-0.003

Table 9: DAPT advantage in the proficiency-based cross-dataset transfer ablation. Each value compares the transfer-induced performance change of a BERT model adapted on a given EFCAMDAT subset against the corresponding non-adapted BERT baseline. Positive values indicate a more favorable transfer outcome for the adapted model. RMSE is treated inversely. Sp.=Spearman; Pe.=Pearson.

Table 10: Full results for the proficiency-based cross-dataset transfer ablation. Values in parentheses show the difference from the corresponding model's in-domain result.

Pretraining data	n	RMSE ↓	QWK ↑	Spearman ↑	Pearson ↑
IELTS → FCE					
Full EFCAMDAT	In-domain	2.101	0.598	0.652	0.678
	50	2.557 (+0.456)	0.429 (-0.169)	0.490 (-0.162)	0.515 (-0.163)
	100	2.433 (+0.332)	0.483 (-0.115)	0.596 (-0.056)	0.616 (-0.062)
	200	2.271 (+0.170)	0.510 (-0.088)	0.590 (-0.062)	0.605 (-0.073)
A1–A2 only	In-domain	2.177	0.581	0.647	0.665
	50	2.502 (+0.325)	0.322 (-0.259)	0.458 (-0.189)	0.457 (-0.208)
	100	2.390 (+0.213)	0.502 (-0.079)	0.574 (-0.073)	0.595 (-0.070)
	200	2.299 (+0.122)	0.509 (-0.072)	0.569 (-0.078)	0.579 (-0.086)
B1–B2 only	In-domain	2.064	0.609	0.662	0.679
	50	2.489 (+0.425)	0.383 (-0.226)	0.474 (-0.188)	0.476 (-0.203)
	100	2.388 (+0.324)	0.415 (-0.194)	0.541 (-0.121)	0.546 (-0.133)
	200	2.294 (+0.230)	0.542 (-0.067)	0.618 (-0.044)	0.628 (-0.051)
B2–C1 only	In-domain	2.143	0.550	0.636	0.653
	50	2.472 (+0.329)	0.388 (-0.162)	0.465 (-0.171)	0.481 (-0.172)
	100	2.489 (+0.346)	0.478 (-0.072)	0.566 (-0.070)	0.590 (-0.063)
	200	2.192 (+0.049)	0.567 (+0.017)	0.619 (-0.017)	0.632 (-0.021)
No EFCAMDAT	In-domain	2.316	0.571	0.677	0.684
	50	2.403 (+0.087)	0.417 (-0.154)	0.528 (-0.149)	0.533 (-0.151)
	100	2.253 (-0.063)	0.514 (-0.057)	0.590 (-0.087)	0.597 (-0.087)
	200	2.216 (-0.100)	0.586 (+0.015)	0.610 (-0.067)	0.623 (-0.061)
FCE → IELTS					
Full EFCAMDAT	In-domain	0.737	0.689	0.764	0.778
	50	0.832 (+0.095)	0.375 (-0.314)	0.699 (-0.065)	0.699 (-0.079)
	100	0.743 (+0.006)	0.628 (-0.061)	0.727 (-0.037)	0.742 (-0.036)
	200	0.737 (+0.000)	0.660 (-0.029)	0.748 (-0.016)	0.755 (-0.023)
A1–A2 only	In-domain	0.735	0.687	0.795	0.793
	50	0.816 (+0.081)	0.455 (-0.232)	0.698 (-0.097)	0.692 (-0.101)
	100	0.645 (-0.090)	0.723 (+0.036)	0.752 (-0.043)	0.766 (-0.027)
	200	0.648 (-0.087)	0.721 (+0.034)	0.736 (-0.059)	0.765 (-0.028)
B1–B2 only	In-domain	0.758	0.663	0.772	0.777
	50	0.827 (+0.069)	0.360 (-0.303)	0.678 (-0.094)	0.679 (-0.098)
	100	0.706 (-0.052)	0.680 (+0.017)	0.747 (-0.025)	0.744 (-0.033)
	200	0.660 (-0.098)	0.704 (+0.041)	0.746 (-0.026)	0.762 (-0.015)
B2–C1 only	In-domain	0.695	0.627	0.725	0.751
	50	0.812 (+0.117)	0.405 (-0.222)	0.692 (-0.033)	0.690 (-0.061)
	100	0.660 (-0.035)	0.715 (+0.088)	0.753 (+0.028)	0.772 (+0.021)
	200	0.652 (-0.043)	0.762 (+0.135)	0.745 (+0.020)	0.765 (+0.014)
No EFCAMDAT	In-domain	0.632	0.712	0.784	0.785
	50	0.661 (+0.029)	0.722 (+0.010)	0.751 (-0.033)	0.749 (-0.036)
	100	0.648 (+0.016)	0.733 (+0.021)	0.767 (-0.017)	0.772 (-0.013)
	200	0.607 (-0.025)	0.808 (+0.096)	0.793 (+0.009)	0.802 (+0.017)

For RMSE, lower is better; therefore, a positive difference from in-domain means performance worsened, while a negative difference means performance improved. For QWK, Spearman, and Pearson, higher is better.